A Multi-Linked Wheeled Robot Inspired by Serpentine Motion

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Research Objectives

Conventional snake-like robots face mobility limitations due to rigid trajectories. This research aims to overcome these constraints by developing a semi-autonomous multi-linked robot with a hybrid actuation system. DC motors provide high-torque propulsion, while paired servo motors enable multi-degree-of-freedom articulation. By implementing a "Follow-the-Leader" time-delay algorithm, the system enables the robot to autonomously adapt its shape to the terrain.

Experimental Conditions

- ✓ **Test Environments:** Experiments were conducted on **flat indoor surfaces** to verify trajectory tracking accuracy and on **standard staircases** to evaluate climbing traction.
- ✓ **Operational Constraints:** Operational stability was secured by limiting servo rotation to **±45°** and capping DC motor velocity to minimize inertial instability during maneuvers.

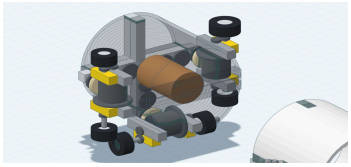
Experimental Methods

- ✓ **Serpentine Gait :** Sinusoidal wave equations (**θ_H**) are applied to servo joints to generate continuous 'S' curve undulation.
- ✓ **Control Algorithm :** Subsequent segments replicate the head's motion after a calculated delay (**Δt**) to create a virtual trace.

Experimental Setup

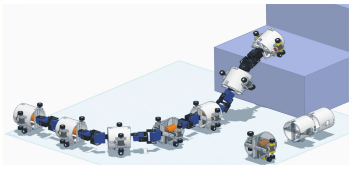
Design of the Experimental Setup

Components



- 3 pairs of wheels driving by DC motors rated at 375 RPM.
- Body angles controlled by 2 pairs of servo motors in a universal joint configuration, allowing multi-DOF movement.

Mechanical Design



- Composed of 8 body segments interconnected by 7 active joints.
- Hybrid control system where servo motors regulate joint articulation and DC motors drive the body for propulsion

Implementation of the Motion Control System

'S' Wave on Flat Surfaces

For the n-th robot joint at time t, the angle are calculated as follows:

$$\theta_H(n,t) = \beta_h + A_h \cdot \sin(\Omega_h \cdot n + \omega_h \cdot t)$$

$$\theta_V(n,t) = 0$$

θ_H and θ_V denote the angles of the horizontal and vertical servo motors, respectively.

Stair Climbing

Trajectory modeling

The circular arc:

$$y = y_c + \sqrt{r^2 - (x - x_c)^2}$$

Joint Angle Calculation

The target bending angle:

$$\theta = 2\arcsin(\frac{db}{a^2 + b^2})$$



Limitations of Pre-Calculated Trajectories

- Restricted movement and limited adaptability to the terrain.

Proposed Method

- A semi-autonomous where only the first join is controlled via a joystick.
- Python processing the inputs and signaling the servo controller.
- The first joint leading the rest with a specific time delay

Experimental Results

Trajectory Tracking Efficiency

- ✓ **Trace Accuracy:** On flat surfaces, the "Follow-the-Leader" algorithm successfully minimized the lateral deviation between the head and tail segments. The body segments followed the generated 'S' curve trajectory smoothly without significant drift.
- ✓ **Motion Fluidity:** Unlike rigid pre-programmed paths, the time-delay control allowed the robot to adjust its shape dynamically based on the operator's real-time input, demonstrating effective semi-autonomous navigation.

Stair Climbing Performance

- ✓ **Torque & Traction:** The 375 RPM DC motors provided sufficient torque to lift the robot's total weight against gravity. The multi-linked structure effectively distributed traction across multiple wheel pairs, preventing slippage on the stair risers.
- ✓ **Geometric Adaptation:** The universal joint configuration allowed the robot to conform to the physical geometry of the stairs. As the robot climbed, the joints naturally adjusted to the height differences, maintaining maximum wheel-to-ground contact.

Discussion

Cause of Error

- ✓ **Using a single PWM signal controlled all DC motors uniformly → deviation from real snake motion where each point's velocity is different**
- ✓ **Axes of the two servo motors (vertical & horizontal) were not perfectly aligned**
- ✓ **Difficult to apply delay accurately since delay time depended on wheel speed which varied along the path**

Improvements Made

- ✓ **Limiting the velocity and acceleration → body stability maintained, load on servo motors reduced**
- ✓ **Rotation range reduced to from 180° → 45° to improve movement**
- ✓ **Applying delay control → robot could follow specified trajectory accurately**

Future Improvement Plan

- ✓ **Difficult to control the motion in curved paths using only the inverse relationship between delay time and speed**
- ✓ **Calculating the curvature of the path and controlling each joint individually could enable more precise movement**

Conclusion

The study confirms that a hybrid wheeled-snake robot using a time-delay control strategy can effectively navigate both flat terrains and vertical obstacles. Future improvements will focus on integrating real-time curvature calculation to control each joint individually, enabling more precise navigation in confined spaces.